Day 031 | Unit 03 | Teacher plan

Integers and Rational Numbers

Represent Signed Quantities and Zero

Standards: 6.NS.C.5

Learning target: I can use positive numbers, negative numbers, and zero in context.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 22

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01 / Learn

Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

Worked example 1

A marker is 32 m below sea level. Write its signed elevation and explain what elevation 0 means.

1. Choose a reference point for zero.

2. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative.

3. Sea level is the reference.

4. Below is negative and above is positive.

5. State the result in the form requested: -32 m; 0 means at sea level.

Answer: -32 m; 0 means at sea level

Worked example 2

A sensor is 7 meters below a reference point. Select all correct interpretations.
A. Its signed position is -7 m.
B. Its signed position is +7 m.
C. The opposite position is 7 m below the reference.
D. A reading of 0 m means at the reference point.

1. Choose a reference point for zero.

2. Below the chosen zero is negative, so the position is -7.

3. Zero identifies the reference

4. the opposite is 7 m above it.

Answer: A, D - Its signed position is -7 m.; A reading of 0 m means at the reference point.

Day 031 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A marker is 14 m above sea level. Write its signed elevation and explain what elevation 0 means.

02 / Constructed response

A marker is 9 m below sea level. Write its signed elevation and explain what elevation 0 means.

03 / Constructed response

A marker is 10 m above sea level. Write its signed elevation and explain what elevation 0 means.

04 / Constructed response

A marker is 10 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

A marker is 35 m above sea level. Write its signed elevation and explain what elevation 0 means.

06 / Constructed response

A marker is 28 m below sea level. Write its signed elevation and explain what elevation 0 means.

07 / Constructed response

A marker is 15 m above sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

08 / Multiple select

A sensor is 2 meters below a reference point. Select all correct interpretations.
A. Its signed position is -2 m.
B. The opposite position is 2 m below the reference.
C. A reading of 0 m means at the reference point.
D. Its signed position is +2 m.

09 / Multiple select

A sensor is 5 meters below a reference point. Select all correct interpretations.
A. Its signed position is +5 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 5 m below the reference.
D. Its signed position is -5 m.

10 / Constructed response

A marker is 15 m below sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

Day 031 | Unit 03 | Answer key

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1. +14 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

2. -9 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

3. +10 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

4. -10 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

5. +35 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

6. -28 m; 0 means at sea level

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

7. +15 m; 0 means at sea level; opposite -15 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response

8. A, C - Its signed position is -2 m.; A reading of 0 m means at the reference point.

Below the chosen zero is negative, so the position is -2. Zero identifies the reference; the opposite is 2 m above it.

6.NS.C.5 | Multiple select

9. B, D - A reading of 0 m means at the reference point.; Its signed position is -5 m.

Below the chosen zero is negative, so the position is -5. Zero identifies the reference; the opposite is 5 m above it.

6.NS.C.5 | Multiple select

10. -15 m; 0 means at sea level; opposite +15 m

Sea level is the reference. Below is negative and above is positive.

6.NS.C.5 | Constructed response